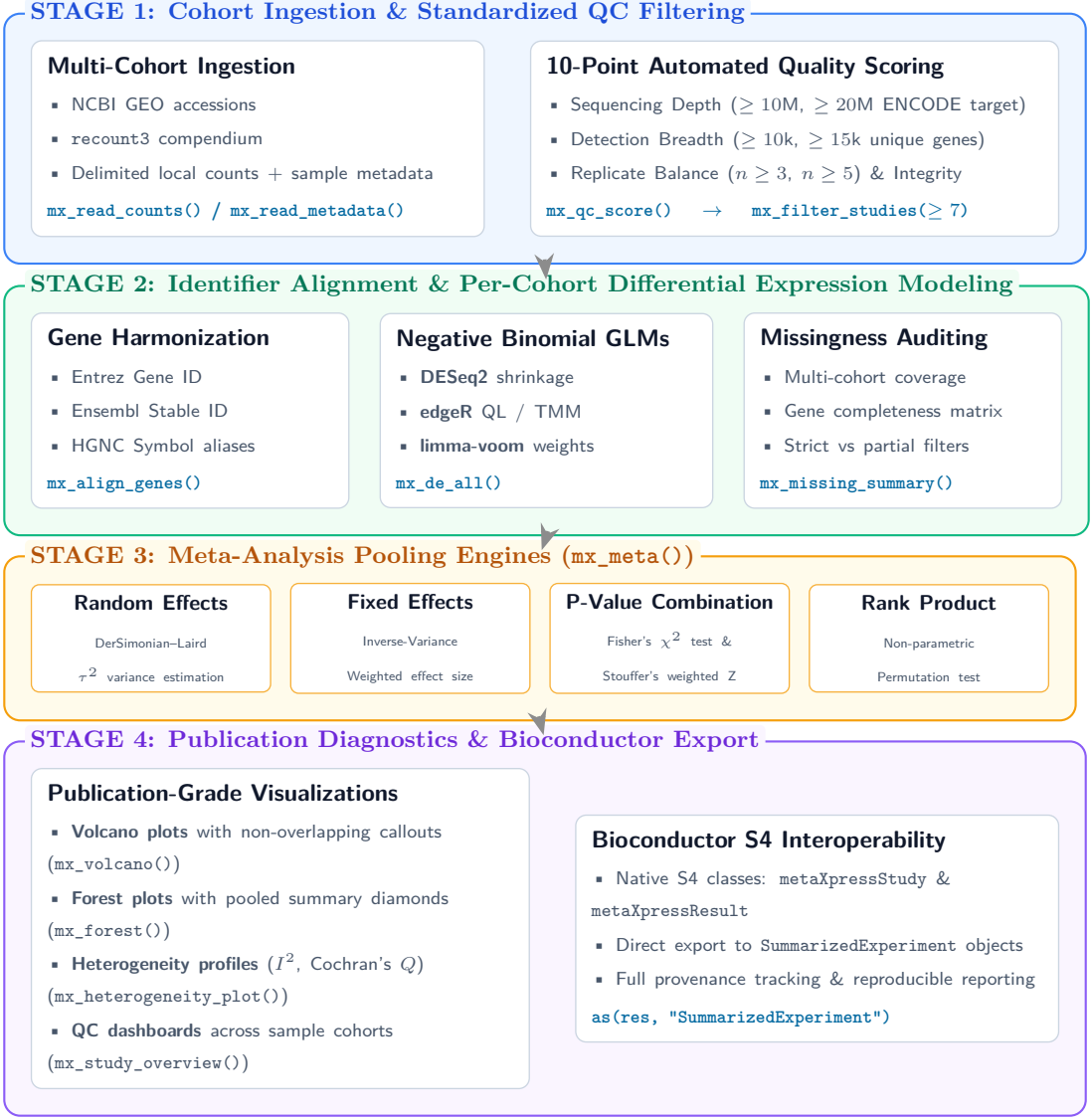

STAGE 1: Cohort Ingestion & Standardized QC Filtering
Multi-Cohort Ingestion
NCBI GEO accessions
recount3 compendium
Delimited local counts + sample metadata
mx_read_counts() / mx_read_metadata()
10-Point Automated Quality Scoring
Sequencing Depth (≥ 10M, ≥ 20M ENCODE target)
Detection Breadth (≥ 10k, ≥ 15k unique genes)
Replicate Balance (n ≥ 3, n ≥ 5) & Integrity
mx_qc_score() → mx_filter_studies(≥ 7)
STAGE 2: Identifier Alignment & Per-Cohort Differential Expression Modeling
Gene Harmonization
Entrez Gene ID
Ensembl Stable ID
HGNC Symbol aliases
mx_align_genes()
Negative Binomial GLMs
DESeq2 shrinkage
edgeR QL / TMM
limma-voom weights
mx_de_all()
Missingness Auditing
Multi-cohort coverage
Gene completeness matrix
Strict vs partial filters
mx_missing_summary()
STAGE 3: Meta-Analysis Pooling Engines (mx_meta())
Random Effects
DerSimonian–Laird
τ² variance estimation
Fixed Effects
Inverse-Variance
Weighted effect size
P-Value Combination
Fisher's χ² test &
Stouffer's weighted Z
Rank Product
Non-parametric
Permutation test
STAGE 4: Publication Diagnostics & Bioconductor Export
Publication-Grade Visualizations
Volcano plots with non-overlapping callouts (mx_volcano())
Forest plots with pooled summary diamonds (mx_forest())
Heterogeneity profiles (I², Cochran's Q) (mx_heterogeneity_plot())
QC dashboards across sample cohorts (mx_study_overview())
Bioconductor S4 Interoperability
Native S4 classes: metaXpressStudy & metaXpressResult
Direct export to SummarizedExperiment objects
Full provenance tracking & reproducible reporting
as(res, "SummarizedExperiment")